Abstractive Text Summarization

Question 3 - From-scratch Transformer Encoder-Decoder

Measured CPU experiment on CNN / Daily Mail 3.0.0. All reported model scores come from autoregressive generation with the validation-selected checkpoint; no pre-trained model weights are used.

1. Data and preprocessing

The official splits contain 287,113 training, 13,368 validation, and 11,490 test pairs. To fit CPU resources, this experiment uses the first 2,048 training pairs and first 1,024 validation pairs in repository order. The training subset is 0.71% of the full training split and is not a random representative sample. All 11,490 test articles are evaluated. No test example trains the tokenizer, model, or hyperparameters.

SentencePiece BPE is trained exclusively on lowercased articles and highlights from the training subset. One shared 4,096-token vocabulary is used for both sides, including PAD=0, UNK=1, BOS=2 and EOS=3. SentencePiece applies NFKC normalization. Source articles are truncated to 400 subwords. Summaries have at most 128 tokens including BOS and EOS (126 content subwords). Batches are padded to their longest sequence. Original reference highlight newlines are retained separately for scoring.

2. Architecture

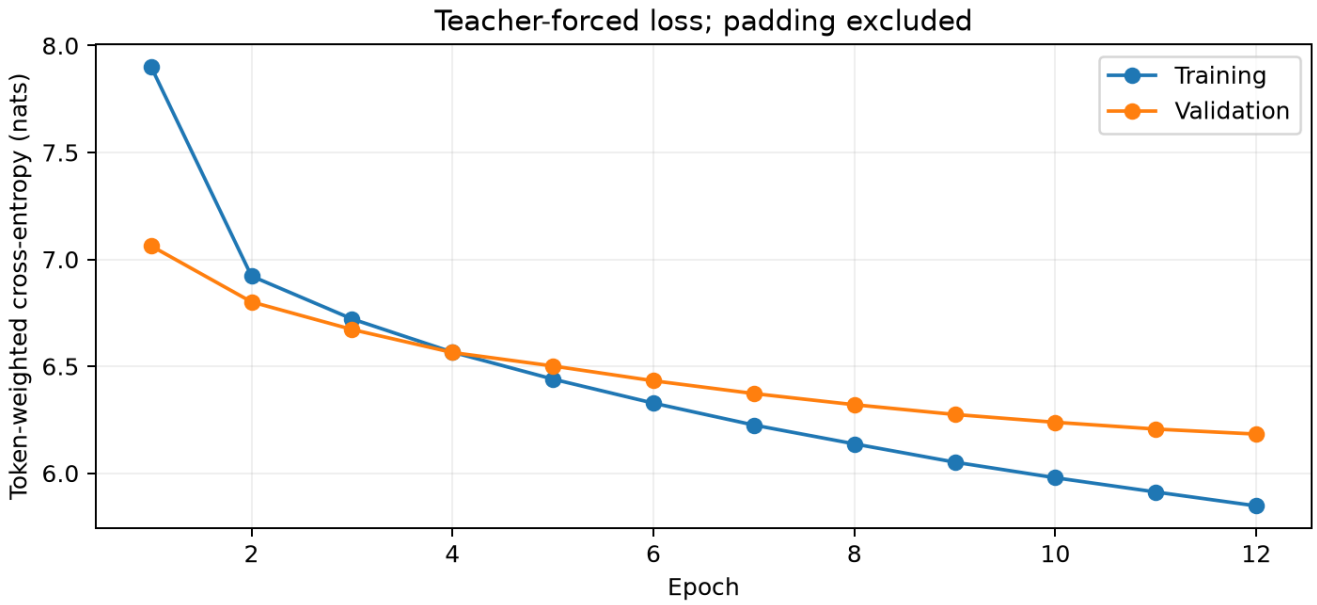
Setting	Value
Encoder / decoder layers	2 / 2
Model / feed-forward width	64 / 256
Attention heads	4 (16 dimensions each)
Dropout / normalization	0.1 / post-residual LayerNorm
Trainable parameters	761,856
Position / output layer	Sinusoidal encoding / linear vocabulary logits

Query, key and value projections are reshaped into heads. Attention is softmax(QK^T / sqrt(16) + mask)V, followed by concatenation and a linear projection. Feed-forward layers use Linear -> ReLU -> Dropout -> Linear. Each sublayer uses LayerNorm(x + Dropout(sublayer(x))). The source padding mask is used in the encoder and cross-attention. Decoder self-attention combines a triangular causal mask and target-key padding mask. All components use elementary tensor operations and basic neural layers.

3. Training and checkpoint selection

Teacher forcing uses gold summaries shifted right as decoder inputs and shifted left as targets. Cross-entropy ignores PAD; epoch losses are weighted by the number of non-padding target tokens. Adam uses betas=(0.9, 0.98), epsilon=1e-9, gradient clipping at 1.0, batch size 16 and seed 42. The learning rate is 0.1 * 64^(-0.5) * min(step^(-0.5), step * 200^(-1.5)). Early stopping has patience 2 after epoch-end validation; the best checkpoint is restored.

The run completed 12 epochs. Best epoch: 12; validation loss: 6.1827 nats/token. Training time: 15.7 minutes on CPU with 4 PyTorch threads. PyTorch 2.14.1+cpu. This width and depth are intentionally smaller than the suggested example configuration because no GPU is available.



Selected epochs are tabulated below; all epochs and learning rates are available in results/history.csv.

Epoch	Training loss	Validation loss
1	7.9018	7.0632
3	6.7215	6.6720
5	6.4398	6.5012
7	6.2250	6.3722
9	6.0514	6.2745
11	5.9118	6.2062
12	5.8469	6.1827

4. Full-test quantitative evaluation

Greedy decoding begins with BOS, feeds predicted tokens back into the decoder, and stops at EOS or 128 generated tokens. PAD and BOS are blocked as output tokens. There is no reference input at generation time. Each article is encoded once; projected decoder keys and values are cached. Caching changes computation cost, not the greedy output (verified by tests).

Scores below are macro averages of per-example F1 multiplied by 100 on all 11,490 test pairs using rouge_score with use_stemmer=True. ROUGE-L uses one sequence; ROUGE-Lsum uses sentence-newline boundaries. Generated text and Lead-3 use an offline punctuation-based sentence heuristic; reference highlights preserve their existing newlines. This heuristic can split abbreviations incorrectly and is a limitation.

System	R-1	R-2	R-L	R-Lsum
Transformer	1.75	0.05	1.63	1.70
Lead-3 (measured)	39.91	17.38	24.90	36.20
Lead-3 (assignment)	40.06	17.48	25.02	36.36

Lead-3 copies the first three sentences of each full original article. It has access to the full article, whereas the model sees only its first 400 subwords. Its measured scores may differ from the assignment reference because sentence segmentation is heuristic. Both systems are scored with the identical local metric configuration.

Transformer minus measured Lead-3 ROUGE-1: -38.16 points. Learning fluent, factual generation from only 2,048 examples is much harder than copying informative news leads. The 4,096-piece vocabulary also splits rare names and numbers into many fragments, and source truncation can remove important details.

There are 4,190 distinct decoded predictions across the 11,490 inputs. Distinct outputs do not imply good summaries: repetition, omitted facts and unsupported statements must still be assessed. The original run collapsed to one output. The revised run reduces the learning-rate scale from 1.0 to 0.1 following validation-only diagnostics. The model architecture, training subset, tokenizer and seed are unchanged. The checkpoint is selected by validation loss, before final test scoring. The original test set has already been evaluated, so this is a reused benchmark, not a new blind held-out experiment. No test metric selects the revised hyperparameters or checkpoint.

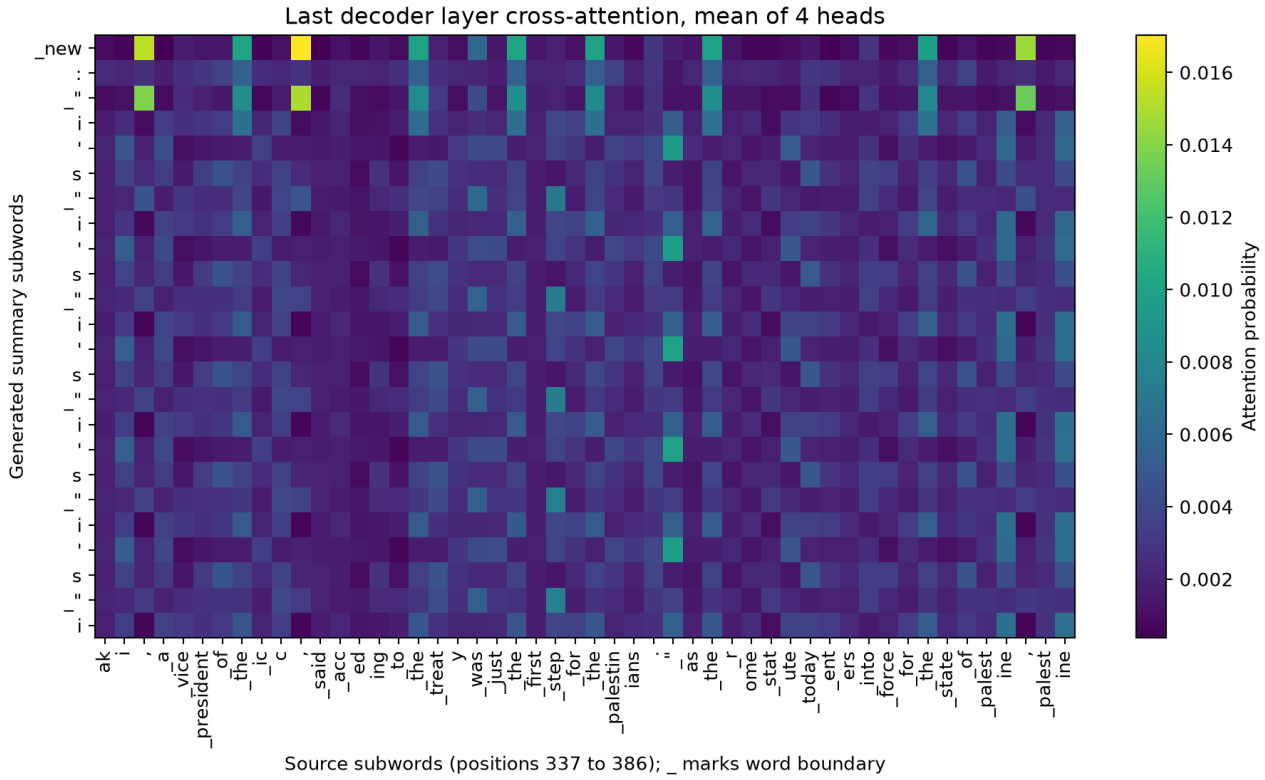
EOS-terminated outputs: 3,964/11,490. Outputs with any repeated whitespace trigram: 11,036/11,490. Generation plus scoring took 2.8 minutes. Repetition is measured on unedited decoded outputs; it is not corrected with sampling or repetition penalties.

Interpretation and limits

ROUGE measures reference overlap rather than factual consistency. Low cross-entropy alone cannot establish good summarization because teacher forcing supplies correct preceding words during validation. Greedy generation must recover from its own errors. This is one seed on a small, ordered subset, with a 1,024-example validation subset; no confidence intervals or claim of competitive summarization quality are made. More representative training data, a wider/deeper model, longer training, and additional validation would be the first improvements. Test scores must not guide those choices.

5. Attention recorded during generation

The heatmap uses cross-attention from the last decoder layer for test example 1. Four heads are averaged. Each row was recorded while generating its labelled predicted token; no gold target was fed to the model. The source window contains 50 consecutive subwords starting at index 337, chosen to maximize summed attention over all generated steps. For legibility the figure shows the first 24 generated subwords. Underscores mark SentencePiece word boundaries.



The selected window contains 13.6% of attention mass per generated row on average (over all generated rows). Probabilities are not renormalized after cropping. The complete attention matrix is supplied in results/attention.npz. A large weight indicates model focus, not proof that a source fact caused a correct prediction. The largest individual weight is 0.020997, compared with uniform 1/400 = 0.002500. A uniform 50-token window would hold 12.5% of the mass. Concentration alone does not establish a correct semantic alignment. Repeated rows alongside repeated output can indicate a decoding loop; diffuse attention can indicate weak alignment. Neither pattern alone proves an error mechanism.

6. Qualitative analysis protocol

The next five pages show the first five test examples in fixed repository order. This selection was fixed before evaluating outputs and avoids cherry-picking. Article excerpts are explicitly labelled; references and generated summaries are unedited and complete. A complete per-test prediction file is provided separately without articles. The diagnostics below each example describe repetition and overlap; they do not certify facts.

7.1. Test example 1

Dataset ID: f001ec5c4704938247d27a44948eebb37ae98d01

Article excerpt (first 1,100 characters)

(CNN)The Palestinian Authority officially became the 123rd member of the International Criminal Court on Wednesday, a step that gives the court jurisdiction over alleged crimes in Palestinian territories. The formal accession was marked with a ceremony at The Hague, in the Netherlands, where the court is based. The Palestinians signed the ICC's founding Rome Statute in January, when they also accepted its jurisdiction over alleged crimes committed "in the occupied Palestinian territory, including East Jerusalem, since June 13, 2014." Later that month, the ICC opened a preliminary examination into the situation in Palestinian territories, paving the way for possible war crimes investigations against Israelis. As members of the court, Palestinians may be subject to counter-charges as well. Israel and the United States, neither of which is an ICC member, opposed the Palestinians' efforts to join the body. But Palestinian Foreign Minister Riad al-Malki, speaking at Wednesday's ceremony, said it was a move toward greater justice. "As Palestine formally becomes a State Party to the Rome St

Reference highlights (original newlines)

Membership gives the ICC jurisdiction over alleged crimes committed in Palestinian territories since last June .
Israel and the United States opposed the move, which could open the door to war crimes investigations against Israelis .

Generated summary (unedited greedy output)

new: "i's "i's "i's "i's "i's "i's "i's "i's "i's "i's "i's "i's "i's "i's "i's "i's" "i's" "i's" "i's "i's "i's" "i's "i's "i's" "i's" "i's"

Observed error indicators

The key reference information is ICC jurisdiction and Palestinian membership. Of the diagnostic source-name/topic strings icc, palestin, the output contains none. String presence alone does not establish a correct factual statement.

The unedited output contains 26 whitespace-delimited words; 16 repeated trigram occurrences beyond their first appearances. It shares 0 distinct whitespace tokens with the reference. These counts diagnose repetition and lexical overlap, not factual accuracy. Compare the source excerpt and reference above to check whether the output preserves the event, participants, and quantities. Generic or repeated text does not identify the key facts even when it contains common reference words.

7.2. Test example 2

Dataset ID: 230c522854991d053fe98a718b1defa077a8efef

Article excerpt (first 1,100 characters)

(CNN)Never mind cats having nine lives. A stray pooch in Washington State has used up at least three of her own after being hit by a car, apparently whacked on the head with a hammer in a misguided mercy killing and then buried in a field -- only to survive. That's according to Washington State University, where the dog -- a friendly white-and-black bully breed mix now named Theia -- has been receiving care at the Veterinary Teaching Hospital. Four days after her apparent death, the dog managed to stagger to a nearby farm, dirt-covered and emaciated, where she was found by a worker who took her to a vet for help. She was taken in by Moses Lake, Washington, resident Sara Mellado. "Considering everything that she's been through, she's incredibly gentle and loving," Mellado said, according to WSU News. "She's a true miracle dog and she deserves a good life." Theia is only one year old but the dog's brush with death did not leave her unscathed. She suffered a dislocated jaw, leg injuries and a caved-in sinus cavity -- and still requires surgery to help her breathe. The veterinary hospita

Reference highlights (original newlines)

Theia, a bully breed mix, was apparently hit by a car, whacked with a hammer and buried in a field .
"She's a true miracle dog and she deserves a good life," says Sara Mellado, who is looking for a home for Theia .

Generated summary (unedited greedy output)

new: "i's "i's "i's "i's "i's "i's "i's "i's "i's "i's "i's "i's "i's "i's "i's "i's "i's "i's "i's "i's "i's "i's "i's "i's "i's "i's "i" "i's "i" "i's "i's "we" "i's "

Observed error indicators

The key reference information is Theia surviving severe injuries, her rescue and care. Of the diagnostic source-name/topic strings theia, dog, the output contains none. String presence alone does not establish a correct factual statement.

The unedited output contains 34 whitespace-delimited words; 23 repeated trigram occurrences beyond their first appearances. It shares 0 distinct whitespace tokens with the reference. These counts diagnose repetition and lexical overlap, not factual accuracy. Compare the source excerpt and reference above to check whether the output preserves the event, participants, and quantities. Generic or repeated text does not identify the key facts even when it contains common reference words.

7.3. Test example 3

Dataset ID: 4495ba8f3a340d97a9df1476f8a35502bcce1f69

Article excerpt (first 1,100 characters)

(CNN)If you've been following the news lately, there are certain things you doubtless know about Mohammad Javad Zarif. He is, of course, the Iranian foreign minister. He has been U.S. Secretary of State John Kerry's opposite number in securing a breakthrough in nuclear discussions that could lead to an end to sanctions against Iran -- if the details can be worked out in the coming weeks. And he received a hero's welcome as he arrived in Iran on a sunny Friday morning. "Long live Zarif," crowds chanted as his car rolled slowly down the packed street. You may well have read that he is "polished" and, unusually for one burdened with such weighty issues, "jovial." An Internet search for "Mohammad Javad Zarif" and "jovial" yields thousands of results. He certainly has gone a long way to bring Iran in from the cold and allow it to rejoin the international community. But there are some facts about Zarif that are less well-known. Here are six: . In September 2013, Zarif tweeted "Happy Rosh Hashanah," referring to the Jewish New Year. That prompted Christine Pelosi, the daughter of House Mino

Reference highlights (original newlines)

Mohammad Javad Zarif has spent more time with John Kerry than any other foreign minister .
He once participated in a takeover of the Iranian Consulate in San Francisco .
The Iranian foreign minister tweets in English .

Generated summary (unedited greedy output)

new: "i's "i's "i's "i's "i's "i" "i" "i's "i" "i" "i's "i's "i's "i" "i" "i" "i" obama"

Observed error indicators

The key reference information is Zarif, diplomacy with John Kerry, and English tweets. Of the diagnostic source-name/topic strings zarif, kerry, the output contains none. String presence alone does not establish a correct factual statement.

The unedited output contains 51 whitespace-delimited words; 38 repeated trigram occurrences beyond their first appearances. It shares 0 distinct whitespace tokens with the reference. These counts diagnose repetition and lexical overlap, not factual accuracy. Compare the source excerpt and reference above to check whether the output preserves the event, participants, and quantities. Generic or repeated text does not identify the key facts even when it contains common reference words.

7.4. Test example 4

Dataset ID: a38e72fed88684ec8d60dd5856282e999dc8c0ca

Article excerpt (first 1,100 characters)

(CNN)Five Americans who were monitored for three weeks at an Omaha, Nebraska, hospital after being exposed to Ebola in West Africa have been released, a Nebraska Medicine spokesman said in an email Wednesday. One of the five had a heart-related issue on Saturday and has been discharged but hasn't left the area, Taylor Wilson wrote. The others have already gone home. They were exposed to Ebola in Sierra Leone in March, but none developed the deadly virus. They are clinicians for Partners in Health, a Boston-based aid group. They all had contact with a colleague who was diagnosed with the disease and is being treated at the National Institutes of Health in Bethesda, Maryland. As of Monday, that health care worker is in fair condition. The Centers for Disease Control and Prevention in Atlanta has said the last of 17 patients who were being monitored are expected to be released by Thursday. More than 10,000 people have died in a West African epidemic of Ebola that dates to December 2013, according to the World Health Organization. Almost all the deaths have been in Guinea, Liberia and Si

Reference highlights (original newlines)

17 Americans were exposed to the Ebola virus while in Sierra Leone in March .
Another person was diagnosed with the disease and taken to hospital in Maryland .
National Institutes of Health says the patient is in fair condition after weeks of treatment .

Generated summary (unedited greedy output)

new: new: "i-old" new: "i-old, police say .
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7.5. Test example 5

Dataset ID: c27cf1b136cc270023de959e7ab24638021bc43f

Article excerpt (first 1,100 characters)

(CNN)A Duke student has admitted to hanging a noose made of rope from a tree near a student union, university officials said Thursday. The prestigious private school didn't identify the student, citing federal privacy laws. In a news release, it said the student was no longer on campus and will face student conduct review. The student was identified during an investigation by campus police and the office of student affairs and admitted to placing the noose on the tree early Wednesday, the university said. Officials are still trying to determine if other people were involved. Criminal investigations into the incident are ongoing as well. Students and faculty members marched Wednesday afternoon chanting "We are not afraid. We stand together," after pictures of the noose were passed around on social media. At a forum held on the steps of Duke Chapel, close to where the noose was discovered at 2 a.m., hundreds of people gathered. "You came here for the reason that you want to say with me, 'This is no Duke we will accept. This is no Duke we want. This is not the Duke we're here to experi

Reference highlights (original newlines)

Student is no longer on Duke University campus and will face disciplinary review .
School officials identified student during investigation and the person admitted to hanging the noose, Duke says .
The noose, made of rope, was discovered on campus about 2 a.m.

Generated summary (unedited greedy output)

new: "we, police say .
new: "we, police say .
new: "we, police say .
new: "we, police say .
new: "we, police say .
new: "we, police say .
new: "we, police say .
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new: "we, police say .
new: "we, police say .
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new: "we, police say .
new: "we, police say .
new: "we, police say .

Observed error indicators

The key reference information is Duke student disciplinary review following the noose incident. Of the diagnostic source-name/topic strings duke, noose, the output contains none. String presence alone does not establish a correct factual statement.

The unedited output contains 75 whitespace-delimited words; 68 repeated trigram occurrences beyond their first appearances. It shares 1 distinct whitespace tokens with the reference. These counts diagnose repetition and lexical overlap, not factual accuracy. Compare the source excerpt and reference above to check whether the output preserves the event, participants, and quantities. Generic or repeated text does not identify the key facts even when it contains common reference words.

8. Collapse investigation and correction

The original checkpoint was audited on validation indices 0, 1, 2, 100, 500 and 1000. All six source token sequences were different. Original encoder layer 1 retained differences (RMS 0.86-0.87), but layer 2 reduced them to about 0.0013. Fresh random weights produced encoder differences around 1.36-1.37. This is evidence of learned representation collapse rather than repeated tokenized input.

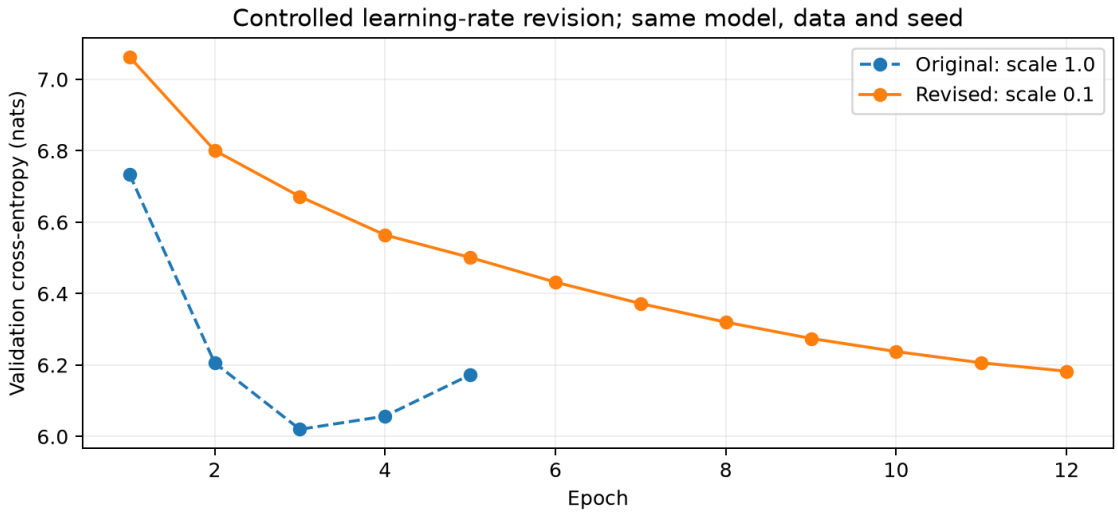
Cached and uncached greedy IDs matched exactly for all six real validation articles, including per-article versus batch runs and a shortened validation source to exercise padding. Hooks checked both decoder layers at every generation step: encoder K/V values and source masks were correct; blocked positions had zero attention. Cached versus full-prefix logit error was at most 5.7e-06. The complete original validation loss recomputed to exactly 6.0200552, matching the checkpoint. No decoder routing or cache defect was found.

The original schedule peaked at 0.0088388, which is aggressive for this small post-normalization model. A controlled retraining uses a 0.1 scale (peak 0.00088388), a maximum of 12 epochs and the same patience of 2. No architecture or decoding changes are introduced to force diverse outputs. The full audit is repeated on the revised validation checkpoint.

Validation diagnostic	Original	Revised
Distinct input sequences / 6	6	6
Distinct generated sequences / 6	1	6
Encoder RMS: article 1 vs 2	0.001306	1.445921
Cached = uncached	True	True
Recomputed validation loss	6.020055	6.182745

The original data/tokenizer reuse was valid, but the scripts failed to check it. The corrected scripts validate counts, unique/disjoint IDs, portable content hashes, BPE vocabulary, special token IDs and tokenizer fingerprints. Fresh streams from pinned revision 96df5e686bee6baa90b8bee7c28b81fa3fa6223d matched all three saved subsets. Training-only tokenizer reconstruction reproduced every piece and score.

9. Stability comparison



The comparison above uses validation only. The original run can have a lower teacher-forced loss while ignoring its source: the correct preceding target words still help next-token prediction. Restoring source sensitivity does not itself guarantee a lower loss, fluent generation or accurate facts. The revised checkpoint is chosen by the lowest validation loss within the revised, fixed training run. The unchanged small data sample remains a major limitation.

10. Verification and reproduction

Unit tests verify future-token isolation, source and target padding masking, attention normalization, ignored padding loss, summary shifting, sinusoidal positions, optimization on a fixed mini-batch, sentence boundaries, and equality of cached and full-prefix greedy predictions. See test_results.txt for the recorded run and README.md for commands.

Observed host: Python 3.14.8 on Windows 11; PyTorch runtime 2.14.1+cpu (distribution metadata 2.14.1), NumPy 2.5.4 in the local dependency overlay, datasets 5.1.0, SentencePiece 0.2.2, rouge-score 0.1.2, matplotlib 3.11.2, reportlab 5.1.0, nltk 3.10.4 and protobuf 7.36.2. The original training run used base NumPy 2.5.3; this revised run uses 2.5.4. PyTorch is unchanged. A separate virtual environment with these pinned direct dependencies and system-site-packages passed pip check, all 14 tests, dataset verification and the offline downloader reuse path. A fully isolated clean-machine installation is not claimed.

The archive includes source scripts, this report, measured metrics, training history, figures, five qualitative examples, test predictions and generation-time attention. It excludes dataset downloads, model checkpoints, the tokenizer training corpus and installed dependencies. Tokenizer and model artifacts are regenerated in a separate work directory.

References

Assignment: NLP Group Assignment 2, DSL 504, Question 3, pages 8-11.

Vaswani et al. (2017). Attention Is All You Need. https://arxiv.org/abs/1706.03762

CNN / Daily Mail 3.0.0, canonical Hugging Face repository: https://huggingface.co/datasets/abisee/cnn_dailymail

SentencePiece: https://github.com/google/sentencepiece

ROUGE implementation: https://github.com/google-research/google-research/tree/master/rouge

PyTorch softmax documentation: https://docs.pytorch.org/docs/stable/generated/torch.nn.Softmax.html